Replication with Causual Forest: 23 US Youth Risk Data Analysis

Step 1: Simply try to replicate the result in the original paper

Summary of the Discrepancy

1. **Physical activity days:** the authors claim that it was measured by "measured by the number of days per week the adolescent was physically active…", however, they reported that the maximum is 4, which should be 7
2. **Numbers of row** (N = 10315) in `df_conf` does not fully matched with the original sample size (N = 10334).
3. **Parental monitoring** was not reverse-coded even though they said so in the method section.
4. Their results for model 2 (male * media use) are inconsistent between their table and their main texts.
5. The logistic regression results are basically replicated. The small differences are likely due to reasons mentioned above.

Data Loading and Cleaning

```
# 1. Pre-process target variables (needs closer look based on 'Data cleaning pipeline for causal forest
df <- df |>
  mutate(
    # 1.0 Generate sequential 5-digit string ID
    id = sprintf("%05d", row_number()),

    # 1.1 Recode Q2 (Sex: 1 = Female, 2 = Male) -> (0 = Female, 1 = Male)
    Q2 = as.numeric(Q2) - 1,

    # 1.3 Recode Q26 (Depressive symptoms: 1 = Yes, 2 = No) -> (1 = Yes, 0 = No)
    Q26 = case_match(
      as.numeric(Q26),
      1 ~ 1,
      2 ~ 0,
      .default = NA_real_
    ),

    # 1.4 Recode Q80 into 3-level categorical/ordered variable:
    # 1 = Low (categories 1-3: non-use to about once a week)
    # 2 = Moderate (categories 4-5: a few times a week to about once a day)
    # 3 = High (categories 6-8: several times a day or more)
    Q80 = case_match(
      as.numeric(Q80),
      1:3 ~ 1,
      4:5 ~ 2,
      6:8 ~ 3,
      .default = NA_real_
    ),

    # 1.5 Convert Q1 (Age categories) into continuous approximate age in years:
```

```
    # Cat 1 (<= 12) -> 12, Cat 2 (13) -> 13, ..., Cat 7 (>= 18) -> 18
    # Mathematical shortcut: as.numeric(Q1) + 11 maps 1:7 to 12:18
    Q1 = case_match(
      as.numeric(Q1),
      1:7 ~ as.numeric(Q1) + 11,
      .default = NA_real_
    ),

    # 1.6 Recode Q25 (Electronic bullying: 1 = Yes, 2 = No) -> (1 = Yes, 0 = No)
    Q25 = case_match(
      as.numeric(Q25),
      1 ~ 1,
      2 ~ 0,
      .default = NA_real_
    ),

    # 1.7 Convert Q76 (Physical activity: 1 = 0 days, 2 = 1 day, ..., 8 = 7 days) -> number of days (0
    # Mathematical shortcut: as.numeric(Q76) - 1 maps 1:8 to 0:7
    Q76 = case_match(
      as.numeric(Q76),
      1:8 ~ as.numeric(Q76) - 1,
      .default = NA_real_
    ),
    # 1.8 Reverse-code School Connectedness (1-5 -> 5-1)
    Q103 = 6 - as.numeric(Q103)
  )

# 2. Confirm Variables
confirm_cols <- c(
  id                   = "id",
  depressive_symptoms  = "Q26",  # DV: depressive symptoms
  social_media_use     = "Q80",  # IV: social media use
  parental_monitoring  = "Q104", # Moderator: parental monitoring (oversight)
  male                 = "Q2",   # Moderator: sex (0 = Female, 1 = Male)
  age                  = "Q1",   # Control: age
  electronic_bullying  = "Q25",  # Control: electronic bullying
  physical_activity    = "Q76",  # Control: physical activity
  school_connectedness = "Q103"  # Control: school connectedness
)

# 3. Explore Variables
explore_cols <- c(
  race_ethnicity       = "Q5",   # Race / ethnicity
  sexual_orientation   = "Q64",  # Sexual orientation
  alcohol_use_30d      = "Q42",  # Alcohol use (past 30 days)
  parent_emotion_abuse = "Q89",  # Parental emotional abuse
  parent_phys_abuse    = "Q90",  # Parental physical abuse
  adult_meet_needs     = "Q99"   # Adults meeting basic needs
)

# 4. Merge column definitions
final_cols <- c(confirm_cols, explore_cols)
```

```r
# 5. Extract subsets
df_conf <- df |>
  select(all_of(confirm_cols))

df_exp <- df |>
  select(all_of(final_cols))
```

Quick Overview

```r
df_exp |> count(depressive_symptoms)
```

```
## # A tibble: 3 x 2
##   depressive_symptoms     n
##                 <dbl> <int>
## 1                   0 11755
## 2                   1  8108
## 3                  NA   240
```

```r
skim(df_conf)
```

Table 1: Data summary

Name	df_conf
Number of rows	20103
Number of columns	9
Column type frequency:	
character	1
numeric	8
Group variables	None

Variable type: character

skim_variable	n_missing	complete_rate	min	max	empty	n_unique	whitespace
id	0	1	5	5	0	20103	0

Variable type: numeric

skim_variable	n_missing	complete_rate	mean	sd	p0	p25	p50	p75	p100	hist
depressive_symptoms	240	0.99	0.41	0.49	0	0	0	1	1	
social_media_use	4900	0.76	2.67	0.67	1	3	3	3	3	
parental_monitoring	9253	0.54	4.21	1.00	1	4	4	5	5	
male	158	0.99	0.50	0.50	0	0	1	1	1	
age	98	1.00	15.89	1.22	12	15	16	17	18	
electronic_bullying	205	0.99	0.17	0.38	0	0	0	0	1	
physical_activity	1227	0.94	3.98	2.50	0	2	4	7	7	
school_connectedness	8926	0.56	3.38	1.14	1	3	4	4	5	

Missing Values

Note: The original author handled missing values by listwise deletion (see argument in their paper)

```
# 1. Check original sample size
nrow(df)
```

```
## [1] 20103
```

```
# 2. Listwise deletion for df_confirm and check sample size
df_conf <- df_conf |> drop_na()
nrow(df_conf)
```

```
## [1] 10315
```

```
# 3. Listwise deletion for df_explore and check sample size
df_exp <- df_exp |> drop_na()
nrow(df_exp)
```

```
## [1] 9208
```

Replication Results

original paper: https://doi.org/10.1007/s11469-025-01566-3

1. number of cases: Numbers of row (N = 10315) in `df_conf` ***does not fully matched*** with the original sample size (N = 10334).
2. try to replicate the descriptive analysis: only `physical activity` is different

```
options(digits = 2)
skim(df_conf)
```

Table 4: Data summary

Name	df_conf
Number of rows	10315
Number of columns	9
Column type frequency:	
character	1
numeric	8
Group variables	None

Variable type: character

skim_variable	n_missing	complete_rate	min	max	empty	n_unique	whitespace
id	0	1	5	5	0	10315	0

Variable type: numeric

skim_variable	n_missing	complete_rate	mean	sd	p0	p25	p50	p75	p100	hist
depressive_symptoms	0	1	0.44	0.50	0	0	0	1	1	
social_media_use	0	1	2.71	0.63	1	3	3	3	3	
parental_monitoring	0	1	4.22	0.98	1	4	4	5	5	

skim_variable	n_missing	complete_rate	mean	sd	p0	p25	p50	p75	p100	hist
male	0	1	0.51	0.50	0	0	1	1	1	
age	0	1	16.06	1.21	12	15	16	17	18	
electronic_bullying	0	1	0.17	0.38	0	0	0	0	1	
physical_activity	0	1	4.00	2.51	0	2	4	7	7	
school_connectedness	0	1	3.39	1.14	1	3	4	4	5	

```
df_conf %>%
  count(social_media_use) %>%
  mutate(
    level = case_match(social_media_use, 1 ~ "Low", 2 ~ "Moderate", 3 ~ "High"),
    percent = percent(n / sum(n), accuracy = 0.1)
  )
```

```
## # A tibble: 3 x 4
##   social_media_use     n level    percent
##              <dbl> <int> <chr>    <chr>
## 1                1   969 Low      9.4%
## 2                2  1071 Moderate 10.4%
## 3                3  8275 High     80.2%
```

Except for the physical activity days, the rest of the results were replicated.

Table 1 Descriptives

Variable	Mean/percentage	SD	Min	Median	Max
Depressive symptoms	0.44	0.50	0.00	0.00	1.00
Parental monitoring	4.22	0.98	1.00	4.00	5.00
School connectedness	3.39	1.14	1.00	4.00	5.00
Physical activity days	1.91	1.05	1.00	2.00	4.00
Age (years)	16.06	1.21	12.00	16.00	18.00
Gender (male)	50.90%				
Electronically bullied	17.30%				
Social media group					
– Low	9.30%				
– Moderate	10.40%				
– High	80.30%				

"Physical activity was included as a protective behavioral factor, measured by the number of days per week the adolescent was physically active for at least 60 min", and the corresponding result from the original Youth Risk dataset codebook. **Hence, it's likely that the authors had some mistake since their max number is 4 instead of 7.**

Q76	During the past 7 days, on how many days were you physically active for a total of at least 60 minutes per day?			
	1	0 days	2,813	15.9
	2	1 day	1,312	7.1
	3	2 days	1,718	9.8
	4	3 days	2,089	10.9
	5	4 days	1,882	10.1
	6	5 days	2,669	13.7
	7	6 days	1,544	8.0
	8	7 days	4,849	24.6
		Missing	1,227	

3. try to replicate the logistic regression results

```r
library(dplyr)
library(knitr)

# 1. Prepare data: low social media use as the reference level
df_model <- df_conf %>%
  mutate(
    social_media_use = factor(
      social_media_use,
      levels = c(1, 2, 3),
      labels = c("Low", "Moderate", "High")
    )
  )

# Fast Wald OR and 95% CI extractor (avoids profile likelihood hang)
format_or <- function(model, target_terms = NULL) {
  cf  <- coef(model)
  se  <- sqrt(diag(vcov(model)))
  pv  <- 2 * (1 - pnorm(abs(cf / se)))

  or  <- exp(cf)
  low <- exp(cf - 1.96 * se)
  upp <- exp(cf + 1.96 * se)

  res <- data.frame(
    Variable = names(cf),
    OR = sprintf("%.2f", or),
    `95% CI` = sprintf("[%.2f, %.2f]", low, upp),
    `p-value` = ifelse(pv < 0.001, "< .001", sprintf("%.3f", pv)),
    check.names = FALSE,
    stringsAsFactors = FALSE
  )

  if (!is.null(target_terms)) {
    res <- res %>% filter(Variable %in% target_terms)
  }
  return(res)
}

# ========================================
# Model 1: Baseline Main Effects
# ========================================
```

```r
m1 <- glm(
  depressive_symptoms ~ social_media_use + male + age +
    electronic_bullying + physical_activity + school_connectedness + parental_monitoring,
  data = df_model,
  family = binomial()
)

# ========================================
# Model 2: Moderation by Gender
# ========================================
m2 <- glm(
  depressive_symptoms ~ social_media_use * male + age +
    electronic_bullying + physical_activity + school_connectedness + parental_monitoring,
  data = df_model,
  family = binomial()
)

# ========================================
# Model 3: Moderation by Parental Monitoring
# ========================================
m3 <- glm(
  depressive_symptoms ~ social_media_use * parental_monitoring + male + age +
    electronic_bullying + physical_activity + school_connectedness,
  data = df_model,
  family = binomial()
)
```

Table 1: Model 1 Baseline Results (Replication)

results were generally replicated: Originally, adolescents who reported moderate use had 1.31 times greater odds of depressive symptoms (95% CI [1.07, 1.60], $p < .01$), while those with high use had 1.77 times greater odds (95% CI [1.52, 2.08], $p < .001$).

Table 7: Model 1: Baseline Logistic Regression Predictors of Depressive Symptoms

	Variable	OR	95% CI	p-value
(Intercept)	(Intercept)	5.50	[2.87, 10.56]	< .001
social_media_useModerate	social_media_useModerate	1.32	[1.08, 1.61]	0.008
social_media_useHigh	social_media_useHigh	1.76	[1.50, 2.06]	< .001
male	male	0.37	[0.34, 0.40]	< .001
age	age	1.02	[0.98, 1.05]	0.339
electronic_bullying	electronic_bullying	3.57	[3.17, 4.03]	< .001
physical_activity	physical_activity	0.94	[0.93, 0.96]	< .001
school_connectedness	school_connectedness	0.71	[0.68, 0.74]	< .001
parental_monitoring	parental_monitoring	0.78	[0.74, 0.81]	< .001

Table 2: Model 2 Gender Moderation (Replication):

results were mostly replicated: originally, in their table:

- Moderate SM×male — 0.99 [0.66, 1.49];
- High SM×male — 0.86 [0.63, 1.18]

But, inconsistently, in their maintext: Contrary to our hypothesis, these interaction terms were not statistically significant (moderate use × gender: OR = 1.12, 95% CI [0.88, 1.43], p =.36; high use × gender: OR = 0.94, 95% CI [0.78, 1.14], p =.53).

Though the conclusion remain the same, as they are all insignificant.

Table 8: Model 2: Moderation Effect of Gender (SM x Male)

	Variable	OR	95% CI	p-value
social_media_useModerate:male	social_media_useModerate:male	0.95	[0.64, 1.43]	0.822
social_media_useHigh:male	social_media_useHigh:male	0.84	[0.62, 1.15]	0.285

Table 3: Model 3 Parental Monitoring Moderation (Replication)

results were generally replicated: the interaction between moderate social media use and parental monitoring was significant (OR = 0.80, 95% CI [0.68, 0.95], $p < .05$), as was the interaction between high use and parental monitoring (OR = 0.87, 95% CI [0.77, 0.98], $p < .05$).

Table 9: Model 3: Moderation Effect of Parental Monitoring (SM x Monitoring)

	Variable	OR	95% CI	p-value
social_media_useModerate:parental_monitoring	social_media_useModerate:parental_monitoring	0.80	[0.68, 0.95]	0.012
social_media_useHigh:parental_monitoring	social_media_useHigh:parental_monitoring	0.85	[0.75, 0.96]	0.011

Step 2: Replication with causal forests

We first export the data which can mostly replicate their results (descriptive analysis and logistic regression)

```
write_csv(df_conf,'df_conf.csv')
```

Then, we import this data to the heterogeneity app to see if it give similar conclusion.

```
# Install heterogeneity package in R

if (!requireNamespace("remotes", quietly = TRUE)) {
  install.packages("remotes")
}
remotes::install_github("invisibleinfo/heterogeneity")

heterogeneity::run_app()
```

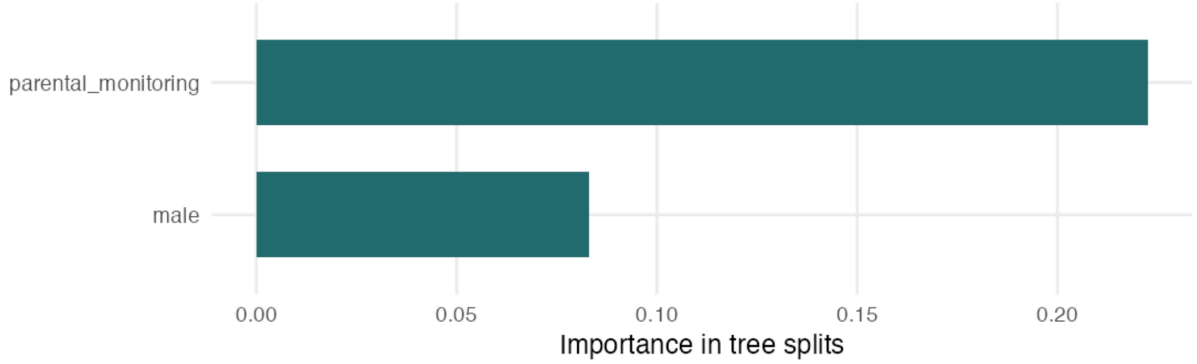
originally, in the paper:

1. higher levels of parental monitoring buffer [social media use —+—> depression]
2. gender does not significantly moderate

from heterogeneity application :

1. the importance seems to be consistent with the original results

Candidate moderators of the treatment effect, ranked by importance



The forest split most often, and earliest, on `parental_monitoring` (0.22) and `male` (0.08).

How to read this plot

These are the leading candidate moderators among your chosen characteristics X as per our exploration with the causal forest. Importance shows which characteristics the estimated treatment effects depend on. Highly correlated characteristics can share importance and should be handled before running the causal forest.

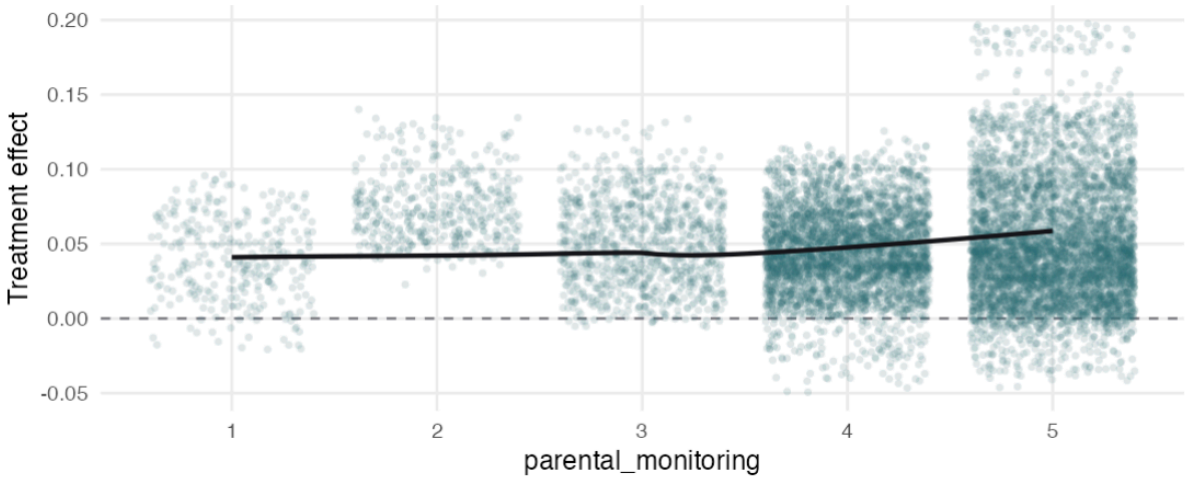
2. the direction seems to be ***inconsistent***

   as in the original paper, low parental monitoring has stronger effect from social media to depression; however, from causal forest, the effect is hightest for high parental monitoring.

1 **parental_monitoring**

SECOND CHARACTERISTIC (OPTIONAL)

None



Following the line, the average treatment effect is lowest at `parental_monitoring` = 1 (0.04) and highest at `parental_monitoring` = 5 (0.06).

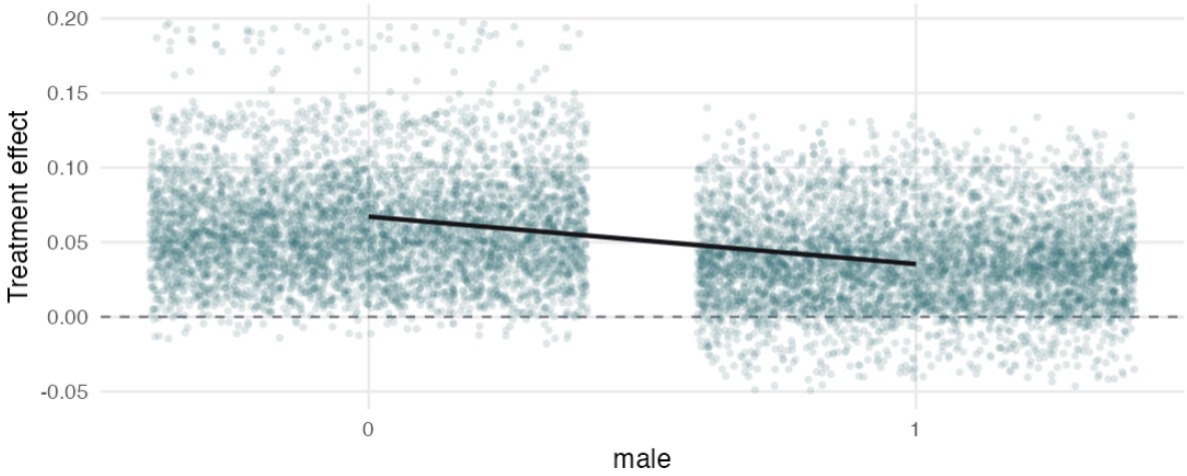
The average treatment effect is 0.05.

---

2 **male**

SECOND CHARACTERISTIC (OPTIONAL)

None



Following the line, the average treatment effect is lowest at `male` = 1 (0.04) and highest at `male` = 0 (0.07).

The average treatment effect is 0.05.